

Project Proposal: The Dynamics of Mastery

Topic: Non-Linear Dynamics and Statistical Physics of Human Skill Acquisition

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Abstract

This project proposes to investigate the statistical laws of human skill acquisition using longitudinal competition data from the World Cube Association (WCA). The study will fit and compare learning-curve models at the individual level, track how solve-time distributions evolve across career phases, and probe the temporal correlation structure of performance fluctuations using Detrended Fluctuation Analysis (DFA). The work will combine non-linear regression, Bayesian model comparison, distribution fitting, and time-series analysis in a reproducible Python pipeline.

Key Definitions:

Solve: A solve refers to a single officially recorded attempt at solving the Rubik's Cube during a World Cube Association (WCA) competition. Each solve is timed, and the resulting duration (in seconds) constitutes one data point in a competitor's longitudinal performance record.

Solver: A solver refers to an individual competitor registered with the WCA who participates in official speedcubing competitions. In this proposal, the term is used to denote a single participant whose sequence of recorded solves forms the basis for individual-level analysis.

Motivation and Research Questions

A foundational finding in cognitive science is that performance improves as a power function of practice, the "Power Law of Practice" (Newell & Rosenbloom, 1981). The idea is simple: early practice yields big gains, later practice yields smaller ones, and there is no hard ceiling. Improvement just keeps slowing down. Heathcote et al. (2000) challenged this. They argued that the power law appears when you average across individuals, but that single individuals actually follow exponential curves.

The distinction matters: under a power law, improvement is scale-free, and performance never truly plateaus. Under an exponential, each person converges toward a fixed limit. Additional practice beyond that limit does essentially nothing. The two models make different predictions about human performance, and distinguishing between them requires individual-level data dense enough to fit curves reliably.

The WCA Results Database is a good fit for this. The World Cube Association runs standardized speedcubing competitions worldwide and publishes every officially recorded solve. I focus on 3×3×3 Rubik's Cube solves, which includes competitors with hundreds of recorded solves spread across multiple years.

RQ1: Learning-Curve Shape: How does the functional form of individual learning curves vary across solvers, and what solver characteristics (career length, practice volume, final skill level) are associated with different model structures?

RQ2: Distributional Evolution: How does the statistical shape of solve-time distributions change across career phases, and which distributional features (skewness, tail weight, variance) shift most between early and late stages?

RQ3: Fluctuation Structure: How does the temporal correlation structure of performance fluctuations differ between early-career vs late-career phases, and what does this reveal about the regularity of skilled performance?

Methodology

The project follows the CRISP-DM lifecycle.

Data

The data source is the publicly available WCA Results Database Export, which contains 14 relational tables. The analysis will be restricted to 3×3×3 solves, filtering for valid results and ordering by competition date to build per-solver longitudinal series. The analysis cohort will comprise solvers with more than 15 competitions, over 200 recorded solves, and a career span of more than 2 years, targeting approximately 30–50 solvers across a range of skill levels to support both individual-level deep analysis and descriptive cross-solver pattern identification.

RQ1 — Learning-Curve Modeling

Three candidate models will be fit per solver:

1. Power law: $t(n) = a \cdot n^{-b} + c$
2. Exponential: $t(n) = a \cdot e^{-b \cdot n} + c$
3. Hybrid: $t(n) = w \cdot a \cdot n^{-b} + (1 - w) \cdot d \cdot e^{-e \cdot n} + c$

where $t(n)$ denotes the solve time for the n -th attempt, and a , b , c , d , e , and w are model parameters. The hybrid model reduces to the pure power-law form when $w = 1$ and to the pure exponential form when $w = 0$.

Initial parameter values will be obtained using non-linear least squares via the `curve_fit` function from the SciPy optimization library (`scipy.optimize.curve_fit`) to improve convergence and reduce the risk of local minima. Model comparison will then be conducted in a Bayesian framework, implemented in PyMC, using weakly informative priors specified a priori and Bayes Factors interpreted following Kass and Raftery (1995). Residual diagnostics will include runs tests and residual plots. Solver-level characteristics including official solve count, career span,

and last-observed average will then be examined descriptively for associations with model preference.

RQ2 — Distributional Analysis

Each solver's career will be split into quartiles by cumulative official solve count, with tercile and quintile splits used as robustness checks. Within each quartile, Gaussian, log-normal, and ex-Gaussian distributions will be fitted. The ex-Gaussian (a convolution of a Gaussian and an exponential component) is standard in reaction-time research (Matzke and Wagenmakers, 2009) and separates the main body of the distribution from slow-tail behavior. Model comparison will use the Akaike Information Criterion (AIC), with Quantile–Quantile (Q–Q) plots and bootstrap-based goodness-of-fit checks for diagnostics.

RQ3 — Fluctuation Analysis

Residuals from the RQ1 learning-curve fits will be analyzed using Detrended Fluctuation Analysis (DFA; Peng et al., 1994). DFA is a method for quantifying the degree of long-range temporal correlations in a time series. It produces a scaling exponent, commonly denoted α (alpha). An α value of approximately 0.5 indicates that the fluctuations behave like uncorrelated random noise (i.e., each performance fluctuation is independent of previous ones). An α value approaching 1.0 suggests the presence of long-range correlations, sometimes called "1/f noise," meaning that a good or bad performance is more likely to be followed by similar deviations over extended stretches.

The scaling exponent will be estimated separately for early-career and late-career phases. This split will only be applied when each phase contains a sufficiently long series of solves to produce a stable and reliable estimate of α . The time series will be indexed by solve number (i.e., the sequential order of attempts) rather than by calendar time. This means that the analysis will capture statistical dependence across consecutive recorded attempts; however, any gaps in calendar time between competitions (e.g., weeks or months without competing) will not be modeled explicitly. This interpretive distinction will be discussed in the final report. Power spectral density computed over solve order will be used as a complementary diagnostic.

Experiment Design and Reproducibility

Model families and comparison criteria are fixed before fitting to prevent post-hoc selection. All code is version-controlled (Git). Tools: Python (Pandas, NumPy, SciPy, Matplotlib, Seaborn, Statsmodels, PyMC).

Expected Results — KPIs, Baselines, and Evaluation

RQ	Success Criterion	Baseline	Metric
RQ1	Clear model discrimination for at least 60% of the eligible cohort	Simple linear trend in solve time over solve number	Bayes Factor >10; residual plots show no systematic structure; runs test shows no strong evidence of residual non-randomness

RQ2	Distributional shifts identified across career phases for at least 50% of the eligible cohort	Gaussian assumed for all phases	AIC-preferred distribution changes between early and late phases, supported by Q-Q plots and bootstrap-based goodness-of-fit checks
RQ3	DFA scaling exponent computed for each eligible phase with confidence intervals; early- vs. late-career comparison conducted where possible	White-noise benchmark $\alpha=0.5$	Scaling exponent reported with confidence intervals; deviation from $\alpha=0.5$ and early/late differences assessed

Domain-Specific Lecture (194.068)

Selected: 166.142 Biology

Key References

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